



Vectors · Mark Scheme

Chapter 10: Vectors. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – 2D vectors

- 1 $|\mathbf{v}| = 13$ B1; $\frac{1}{13}(5\mathbf{i} - 12\mathbf{j})$ B1.
- 2 $8\mathbf{i} + \mathbf{j}$ B1; $\sqrt{65}$ B1.
- 3 $\tan \theta = -\frac{3}{4}$ M1; 36.9° below the positive x -axis (i.e. -36.9°) A1; $\frac{4}{5}\mathbf{i} - \frac{3}{5}\mathbf{j}$ B1.
- 4 (a) $k^2 + 36 = 100$ M1; $k = \pm 8$ A1. (b) $15 \times \frac{1}{5}(3\mathbf{i} + 4\mathbf{j})$ M1; $9\mathbf{i} + 12\mathbf{j}$ A1.
- 5(a) $\mathbf{R} = (4 + p)\mathbf{i} - 2\mathbf{j}$ M1; $\frac{4+p}{3} = \frac{-2}{-1}$ M1; $p = 2$ A1.
- 5(b) $\mathbf{R} = 6\mathbf{i} - 2\mathbf{j}$, $|\mathbf{R}| = \sqrt{40} = 2\sqrt{10}$ N M1 A1.
- 5(c) $\tan \theta = \frac{2}{6}$ M1; 18.4° A1.

Section B – Position vectors and ratio

- 6 $4\mathbf{i} + 6\mathbf{j}$ B1; $\sqrt{52} = 2\sqrt{13}$ B1.
- 7 $\mathbf{m} = \frac{1}{2}(\mathbf{a} + \mathbf{b}) = 6\mathbf{i} + 4\mathbf{j}$ M1 A1; $\overrightarrow{AM} = 3\mathbf{i} + 6\mathbf{j}$, $AM = 3\sqrt{5}$ A1.
- 8 $\overrightarrow{AB} = 8\mathbf{i} - 8\mathbf{j}$ B1; $\mathbf{p} = \mathbf{a} + \frac{1}{4}\overrightarrow{AB}$ M1; $= 3\mathbf{j}$ A1.
- 9 $\overrightarrow{ON} = \frac{2}{3}\mathbf{b}$ B1; (a) $\overrightarrow{AN} = \frac{2}{3}\mathbf{b} - \mathbf{a}$ B1; (b) $\overrightarrow{MN} = \frac{2}{3}\mathbf{b} - \frac{1}{2}(\mathbf{a} + \mathbf{b})$ M1; $= -\frac{1}{2}\mathbf{a} + \frac{1}{6}\mathbf{b}$ A1.
- 10(a) $\overrightarrow{OP} = \frac{2}{3}\mathbf{a}$, $\overrightarrow{OQ} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$ M1; $\overrightarrow{PQ} = -\frac{1}{6}\mathbf{a} + \frac{1}{2}\mathbf{b}$ A1.
- 10(b) $\overrightarrow{PR} = k\mathbf{b} - \frac{2}{3}\mathbf{a}$ B1; $\overrightarrow{PR} = \lambda \overrightarrow{PQ}$ M1; compare $\mathbf{a}$: $-\frac{2}{3} = -\frac{\lambda}{6} \Rightarrow \lambda = 4$ A1; compare $\mathbf{b}$: $k = \frac{\lambda}{2} = 2$ A1.
- 10(c) $\overrightarrow{PR} = 4\overrightarrow{PQ}$ M1; $PQ : QR = 1 : 3$ A1.

Section C – Vector geometry

- 11 $\mathbf{w} = -\frac{3}{2}\mathbf{u}$ M1; so parallel A1.
- 12 $\overrightarrow{AB} = 3\mathbf{i} + \mathbf{j}$, $\overrightarrow{BC} = 6\mathbf{i} + 2\mathbf{j}$ M1; $\overrightarrow{BC} = 2\overrightarrow{AB}$ A1; parallel with common point B , so collinear A1.
- 13 $\overrightarrow{AB} = 2\mathbf{i} + (k-2)\mathbf{j}$, $\overrightarrow{AC} = 6\mathbf{i} + 9\mathbf{j}$ M1; $\frac{k-2}{2} = \frac{9}{6}$ M1; $k = 5$ A1.
- 14 (a) $\overrightarrow{AB} = 4\mathbf{i} + \mathbf{j} = \overrightarrow{DC}$ M1 A1; opposite sides equal and parallel, so parallelogram A1. (b) $|AB| = \sqrt{17}$, $|AD| = \sqrt{20}$, not equal, so not a rhombus B1.
- 15(a) $\overrightarrow{OB} = \mathbf{a} + \mathbf{c}$, midpoint $\frac{1}{2}(\mathbf{a} + \mathbf{c})$ M1; midpoint of AC is $\mathbf{a} + \frac{1}{2}(\mathbf{c} - \mathbf{a}) = \frac{1}{2}(\mathbf{a} + \mathbf{c})$ M1; same point, so diagonals bisect each other A1.
- 15(b) $\overrightarrow{OM} = \mathbf{a} + \frac{1}{2}\mathbf{c}$ B1; $\overrightarrow{OX} = s\mathbf{a} + \frac{s}{2}\mathbf{c}$ and $\overrightarrow{OX} = \mathbf{a} + t(\mathbf{c} - \mathbf{a})$ M1; compare: $s = 1 - t$, $\frac{s}{2} = t$ M1; $s = \frac{2}{3}$ A1; $OX : XM = 2 : 1$ A1.

Section D – 3D and scalar product

- 16 $|\mathbf{a}| = 3$ B1; $\frac{1}{3}(2\mathbf{i} - \mathbf{j} + 2\mathbf{k})$ B1.
- 17 $\overrightarrow{AB} = 3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$ B1; $\sqrt{9+4+36}$ M1 $= 7$ A1.
- 18 (a) $3 - 4 + 10 = 9$ M1 A1. (b) $6 - 4 - 2 = 0$, so perpendicular B1.
- 19 Dot product $2 - 2 + 4 = 4$ M1; magnitudes 3 and 3 B1; $\cos \theta = \frac{4}{9}$ M1; $\theta = 63.6^\circ$ A1.



- 20(a)** $\vec{AB} = 3\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ *B1*; $\vec{AC} = \mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ *B1*.
- 20(b)** $\vec{AB} \cdot \vec{AC} = 9$ *M1*; $\cos \theta = \frac{9}{\sqrt{17}\sqrt{30}}$ *M1*; $\theta = 66.5^\circ$ *A1*.
- 20(c)** $\frac{1}{2}\sqrt{17}\sqrt{30}\sin 66.5^\circ$ *M1*; $= 10.4$ *A1*. (Exact: $\frac{1}{2}\sqrt{429}$.)
- 20(d)** $3k + 2 + 2 = 0$ *M1*; $k = -\frac{4}{3}$ *A1*.

Section E – Mechanics

- 21** $\mathbf{v} = 2t\mathbf{i} + 3\mathbf{j}$ *M1*; at $t = 2$: $4\mathbf{i} + 3\mathbf{j}$ m s^{-1} *A1*.
- 22** $\mathbf{a} = 6t\mathbf{i} + 4\mathbf{j}$ *M1 A1*; $\mathbf{v}(1) = \mathbf{i} + 4\mathbf{j}$, speed $\sqrt{17}$ m s^{-1} *A1*.
- 23** $\mathbf{v} = 6t^2\mathbf{i} - 2t\mathbf{j}$ *M1*; parallel to $\mathbf{i} - \mathbf{j}$ when $6t^2 = 2t$ *M1*; $t = \frac{1}{3}$ *A1*.
- 24** (a) $\mathbf{v} = (4 - 2t)\mathbf{i} + (3t^2 - 3)\mathbf{j}$ *M1*; $\mathbf{i}$ -component zero at $t = 2$ *A1*; $\mathbf{v} = 9\mathbf{j}$, speed 9 m s^{-1} *A1*. (b) $\mathbf{a} = -2\mathbf{i} + 6t\mathbf{j} = -2\mathbf{i} + 12\mathbf{j}$ m s^{-2} *B1*.
- 25(a)** $\mathbf{v} = 12\mathbf{i} + (20 - 10t)\mathbf{j}$ *M1*; initial speed $\sqrt{544} = 23.3$ m s^{-1} *A1*.
- 25(b)** $20 - 10t = 0 \Rightarrow t = 2$ *M1 A1*; $\mathbf{r} = 24\mathbf{i} + 20\mathbf{j}$ *A1*.
- 25(c)** $20t - 5t^2 = 0 \Rightarrow t = 4$ *M1*; $\mathbf{v} = 12\mathbf{i} - 20\mathbf{j}$, speed 23.3 m s^{-1} *A1*; same as the initial speed, as expected by symmetry of the path (no air resistance) *B1*.

Roar-level challenge

- 26** $C = (c, 0, 0)$; $\vec{CA} = (1 - c)\mathbf{i} + 2\mathbf{k}$, $\vec{CB} = (3 - c)\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ *M1 A1*; $\vec{CA} \cdot \vec{CB} = 0$: $(1 - c)(3 - c) - 4 = 0$ *M1*; $c^2 - 4c - 1 = 0$ *A1*; $C = (2 \pm \sqrt{5}, 0, 0)$ *A1*.